Hecke operators for multiple Eisenstein series

Samyak Parashar
Nagoya University

The 2nd Multiple Eisenstein Series Research Meeting
November 8, 2026 Nagoya

Hecke operators are sums over sublattices

Lattice functions

$$F(\alpha\Gamma) = \alpha^{-k}F(\Gamma) \quad \forall \alpha \in \mathbb{C}^\times$$

$$(T_nF)(\Gamma) = n^{k-1} \sum_{[\Gamma:L]=n} F(L)$$

$$f(\tau) := F(\Lambda_\tau)$$
$$\Lambda_\tau := \mathbb{Z}\tau \oplus \mathbb{Z}$$

Modular forms

$$T_nf(\tau) = n^{k-1} \sum_{ad=n} d^{-k} \sum_{b=0}^{d-1} f\Big(\frac{a\tau+b}{d}\Big)$$

To define Hecke operators on MES we first define multiple Eisenstein series on all lattices in $\mathbb{C}$.

Multiple Eisenstein series on lattices

Definition 1 (MES on lattices)

We order a lattice $\Gamma \subset \mathbb{C}$ by $\lambda \succ \mu :\Longleftrightarrow \lambda - \mu \in \mathbb{H} \cup \mathbb{R}_{>0}$. For $k_1 \geq 3$ and $k_2, \ldots, k_r \geq 2$,

$$\mathbb{G}_{k_1,\ldots,k_r}(\Gamma) = \sum_{\lambda_1 \succ \cdots \succ \lambda_r \succ 0 \text{ in } \Gamma} \frac{1}{\lambda_1^{k_1} \cdots \lambda_r^{k_r}}.$$

- $\vec{k} = (k_1, \cdots, k_r)$ denotes any admissible index and we use $k = wt(\vec{k}) = k_1 + \cdots + k_r$.
- A sublattice carries the restricted order, and scaling by $\rho > 0$ preserves it: $\mathbb{G}_{\vec{k}}(\rho\Gamma) = \rho^{-k}\mathbb{G}_{\vec{k}}(\Gamma),\ \ k = k_1 + \cdots + k_r$.
- $\mathrm{G}_{\vec{k}}(\tau) := \mathbb{G}_{\vec{k}}(\Lambda_\tau)$ are the MES of Gangl-Kaneko-Zagier (2006).
- $\mathcal{E}_k := \left\langle \mathrm{G}_{\vec{k}} :\ k_1 + \cdots + k_r = k,\ k_i \geq 2 \right\rangle_{\mathbb{Q}} \quad \mathcal{E} = \bigoplus_k \mathcal{E}_k,\ \ \mathcal{E}^{\mathbb{C}} := \mathbb{C} \otimes \mathcal{E}.$

Hecke operators

Definition 2 (Hecke operator)

For $n \geq 1$, a weight k and a q-series $F = \sum_{m \geq 0} a_m q^m$,

$$T_n F := \sum_{m \geq 0} \Big(\sum_{d \mid (m,n)} d^{k-1} a_{mn/d^2} \Big) q^m.$$

On modular forms of weight k this is the classical Hecke operator.

Proposition 3 (Hecke relations)

For all $m, n \geq 1$, with $D = 2\pi i \frac{d}{d\tau}$,

$$T_m T_n = \sum_{d \mid (m,n)} d^{k-1} T_{mn/d^2}, \qquad T_n D = n\, D\, T_n, \qquad T_n \, \mathrm{G}_k = \sigma_{k-1}(n) \, \mathrm{G}_k \quad (k \geq 2).$$

The sublattice picture

Theorem 4

For $\vec{k}$ admissible ($k_1 \geq 3$) of weight k and $n \geq 1$,

$$\begin{aligned} T_n \, \mathrm{G}_{\vec{k}}(\tau) &= (T_n \mathbb{G}_{\vec{k}})(\Lambda_\tau) \\ &= n^{k-1} \sum_{[\Lambda_\tau : L] = n} \mathbb{G}_{\vec{k}}(L) \\ &= n^{k-1} \sum_{ad=n} d^{-k} \sum_{b=0}^{d-1} \mathrm{G}_{\vec{k}}\Big(\frac{a\tau + b}{d}\Big). \end{aligned}$$

The Hecke operator applied to an MES gives a sum of MES over sublattices.

Coloured multiple Eisenstein series

Definition 5 (Coloured MES)

Let Γ be a lattice and $N \geq 1$. The *colour* of $\lambda \in \Gamma$ is its class in $\Gamma/N\Gamma$. For $\vec{k}$ admissible and colours $c_1, \ldots, c_r \in \Gamma/N\Gamma$,

$$\mathbb{G}_{\vec{k}}^{(c_1,\ldots,c_r)}(\Gamma) := \sum_{\substack{\lambda_1 \succ \cdots \succ \lambda_r \succ 0 \text{ in } \Gamma \\ \lambda_i \in c_i}} \frac{1}{\lambda_1^{k_1} \cdots \lambda_r^{k_r}}, \qquad \mathrm{G}_{\vec{k}}^{(c_1,\ldots,c_r)}(\tau) := \mathbb{G}_{\vec{k}}^{(c_1,\ldots,c_r)}(\Lambda_\tau).$$

On Λ_τ, $m\tau + s \mapsto (m, s)$ identifies $\Lambda_\tau/N\Lambda_\tau = (\mathbb{Z}/N\mathbb{Z})^2$.

- With colours $(0, a_i)$ only, these are the level-N MES of Yuan-Zhao (2015) (cf. Kanno (2025)).
- $\mathcal{E}_k^N := \left\langle \mathrm{G}_{\vec{k}}^{(c_1,\ldots,c_r)} : \ \vec{k} \text{ admissible of weight } k, \ c_i \in (\mathbb{Z}/N\mathbb{Z})^2 \right\rangle_{\mathbb{Q}}$.
- $\mathcal{E}_k^1$ is spanned by the admissible $\mathrm{G}_{\vec{k}}$, and $\mathcal{E}_k^N \subseteq \mathcal{E}_k^{NM}$ (refine the colours).

Hecke operators raise the level

Theorem 6 (Hecke operators raise the level)

For all $N, m \geq 1$: $T_m \, \mathcal{E}_k^N \subseteq \mathcal{E}_k^{mN}$.

Explicitly, for $\vec{k}$ admissible of weight k and p prime,

$$T_p \, \mathrm{G}_{\vec{k}} = \left(1 + p^{k-1}\right) \mathrm{G}_{\vec{k}} - p^{k-1} \sum_{\langle c_1, \ldots, c_r \rangle = \mathbb{F}_p^2} \mathrm{G}_{\vec{k}}^{(c)} \, .$$

Here $(c) = (c_1, \ldots, c_r)$, and $\langle c_1, \ldots, c_r \rangle = \mathbb{F}_p^2$ means the colours span $(\mathbb{Z}/p\mathbb{Z})^2$.

In depth one a single colour never spans $\mathbb{F}_p^2$, so G_k is an eigenform. From depth two on, the full-rank colours appear.

The Hecke operators act on $\bigcup_N \mathcal{E}_k^N$. Problem: What would be the eigenfunctions there ?

Hecke operators do not preserve MES

Theorem 7

We have

$$\{F \in \mathcal{E}_5 : \ T_2 F \in \mathcal{E}_5\} = \mathbb{Q}\ \mathrm{G}_5 \oplus \mathbb{Q}\, D\, \mathrm{G}_3,$$
$$\{F \in \mathcal{E}_6 : \ T_2 F \in \mathcal{E}_6\} = \mathbb{Q}\ \mathrm{G}_6 \oplus \mathbb{Q}\ \mathrm{G}_2\, \mathrm{G}_4 \oplus \mathbb{Q}\ \mathrm{G}_2^3 = \mathbb{Q}[\mathrm{G}_2, \mathrm{G}_4, \mathrm{G}_6]_6.$$

In particular, $T_2\ \mathrm{G}_{3,2} \notin \mathcal{E}_5$ *and* $T_2\ \mathrm{G}_{3,3}, T_2\ \mathrm{G}_{2,4} \notin \mathcal{E}_6$.

Where $T_2\, \mathrm{G}_{3,2}$ goes (Theorem 6, $p = 2$): one colour never spans $\mathbb{F}_2^2$, and in depth two

$$T_2\, \mathrm{G}_{3,2} = 17\ \mathrm{G}_{3,2} - 16 \sum_{\langle c_1, c_2 \rangle = \mathbb{F}_2^2} \mathrm{G}_{3,2}^{(c_1, c_2)},$$

a sum over the six ordered bases of $\mathbb{F}_2^2$: level 2, not level 1.

The modularity conjecture

Definition 8 (Hecke core)

$\mathcal{H}_k := \{F \in \mathcal{E}_k : \ T_n F \in \mathcal{E}_k \text{ for all } n \geq 1\}$, the largest Hecke-stable subspace of $\mathcal{E}_k$.

It contains

$$\mathcal{A}_k := \begin{cases} \mathbb{Q}[\mathrm{G}_2, \mathrm{G}_4, \mathrm{G}_6]_k & (k \text{ even}), \\ \left\langle D^r \, \mathrm{G}_j : \ j \geq 3 \text{ odd}, \ j + 2r = k \right\rangle_{\mathbb{Q}} & (k \text{ odd}), \end{cases}$$

since T_n preserves quasimodular forms, and $T_n D^r \, \mathrm{G}_j = n^r \sigma_{j-1}(n) \, D^r \, \mathrm{G}_j$.

Conjecture 9 (Modularity conjecture)

For all $k \geq 2$: (i) $\mathcal{H}_k = \mathcal{A}_k$; (ii) already $\{F \in \mathcal{E}_k : \ T_2 F \in \mathcal{E}_k\} = \mathcal{A}_k$. (ii) implies (i).

In even weight: an MES is quasimodular $\iff$ all of its Hecke images are again MES.

Hecke operators detect quasimodularity

Example 10 (Testing the conjecture for weight 5 and 6)

- Weight 6: $F = \mathrm{G}_{2,4} + 2\,\mathrm{G}_{3,3}$ has

$$T_2 F = 18\,F + \tfrac{65}{4}\,\mathrm{G}_6 \;\in \mathcal{E}_6.$$

  By Theorem 7, F is quasimodular. Indeed $F = \mathrm{G}_2\,\mathrm{G}_4 - \tfrac{2}{3}\,\mathrm{G}_6$.

- Weight 5: $F = 2\,\mathrm{G}_{3,2} + 3\,\mathrm{G}_{2,3}$ has

$$T_2 F = \tfrac{35}{2}\,\mathrm{G}_5 + 20\,\mathrm{G}_{3,2} + 30\,\mathrm{G}_{2,3} \;\in \mathcal{E}_5.$$

  By Theorem 7, $F \in \mathcal{A}_5$. Indeed $F = \tfrac{1}{2}\big(5\,\mathrm{G}_5 - D\,\mathrm{G}_3\big)$.

Quasimodularity is read off from a single Hecke image. T_2 moves $\mathcal{E}_k^1$ into $\mathcal{E}_k^2$ (Theorem 6). Conjecturally it comes back to level 1 only on $\mathcal{A}_k$.

Evidence: the Hecke-stable part is tiny

k	2	3	4	5	6	7	8	9	10	11	12
$\dim_{\mathbb{Q}} \mathcal{E}_k$	1	1	2	3	4	7	9	15	21	32	47
$\dim_{\mathbb{Q}} \mathcal{A}_k$	1	1	2	2	3	3	4	4	5	5	7

- $k \leq 12$: Conjecture 9 is true.

Summary

	Modular forms	Multiple Eisenstein series
Hecke operators	sums over sublattices	sums over sublattices (Theorem 4)
Hecke images	stay in $\mathcal{M}_k$	level $N \to$ level mN (Theorems 6, 7)
Hecke-stable part	all of $\mathcal{M}_k$	$\mathcal{A}_k$? (Conjecture 9)

Future work : Conjecture 9 in general; the Hecke eigenfunctions on $\bigcup_N \mathcal{E}_k^N$; $\mathfrak{sl}_2$ action on $\mathcal{E}^N$.

Thank you!

Backup: compatibility with $\mathfrak{sl}_2$

Theorem 11 ($\mathfrak{sl}_2$-compatibility)

T_n commutes with W, $T_n D = n\, D\, T_n$, and if $T_n F \in \mathcal{E}_k$ then $\delta\, T_n F = n\, T_n\, \delta F$. Hence $\bigoplus_k \mathcal{H}_k$ is an $\mathfrak{sl}_2$-submodule of $\mathcal{E}$.

Here $\mathfrak{sl}_2 = \langle W, D, \delta \rangle$ is the one of Bachmann-Kanno (2026).

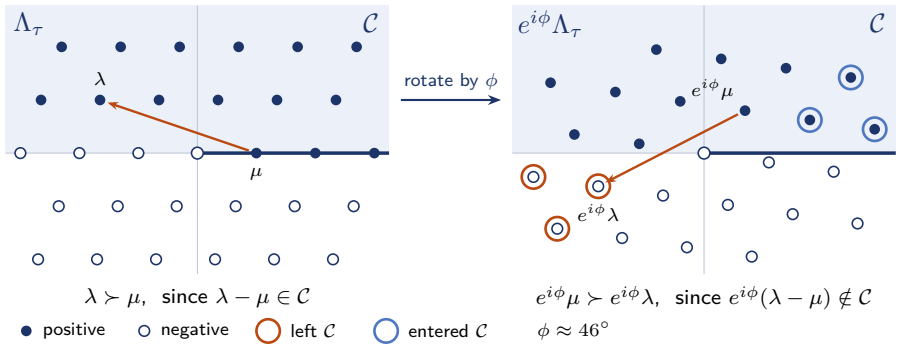
Proposition 12 (Primitive form)

Conjecture 9(ii) holds for all k if and only if, for all k,

$$\{F \in \mathcal{E}_k^1 :\ T_2 F \in \mathcal{E}_k\} = \mathcal{P}_k := \begin{cases} \mathbb{Q}[\mathrm{G}_4, \mathrm{G}_6]_k & (k \text{ even}), \\ \mathbb{Q}\ \mathrm{G}_k & (k \text{ odd}). \end{cases}$$

T_2 moves $\mathcal{E}_k^1$ into $\mathcal{E}_k^2$ (Theorem 6). Conjecturally it comes back to level 1 only on $\mathcal{P}_k$.

Backup: only positive scalings preserve the order



Rotating a lattice changes its order, so $\mathbb{G}_{\vec{k}}$ is homogeneous only under $\rho \in \mathbb{R}_{>0}$.

Backup: lattices up to positive scaling

Definition 13 (Lattice functions of weight k)

$\mathcal{F}_k := \{F\colon \mathcal{R} \to \mathbb{C} \mid F(\rho\Gamma) = \rho^{-k}F(\Gamma) \text{ for all } \rho > 0\}$, $\mathcal{R}$ the set of lattices in $\mathbb{C}$. By Definition 1, $\mathbb{G}_{\vec{k}} \in \mathcal{F}_k$.

Proposition 14 (Real homothety)

Lattices up to positive scaling form

$$\mathcal{R}/\mathbb{R}_{>0} \;\cong\; \mathrm{PSL}_2(\mathbb{R})/\,\mathrm{PSL}_2(\mathbb{Z}) \;\cong\; T^1\big(\mathrm{SL}_2(\mathbb{Z})\backslash\mathbb{H}\big),$$

via $[\Gamma] \leftrightarrow (\tau,\phi)$, $\Gamma \sim (\operatorname{Im}\tau)^{-1/2}e^{i\phi}\Lambda_\tau$*: a point* τ *of the modular surface and a direction* $\phi \in \mathbb{R}/\pi\mathbb{Z}$*. So* $F \in \mathcal{F}_k$ *is a function* $\mathcal{G}_F(\tau,\phi) := F(e^{i\phi}\Lambda_\tau)$ *with*

$$\mathcal{G}_F\big(\gamma\tau,\ \phi + \arg(c\tau+d)\big) = |c\tau+d|^k\,\mathcal{G}_F(\tau,\phi) \qquad (\gamma \in \mathrm{SL}_2(\mathbb{Z})).$$

Backup: MES live on the unit tangent bundle

Proposition 15 (MES as lattice functions)

There is an injective algebra map $\mathrm{Lat}\colon \mathcal{E} \hookrightarrow \bigoplus_k \mathcal{F}_k$ *with*

$$\mathrm{Lat}(\mathrm{G}_{\vec{k}}) = \mathbb{G}_{\vec{k}} \quad (\vec{k} \text{ admissible}) \qquad \text{and} \qquad \mathrm{Lat}(F)(\Lambda_\tau) = F(\tau) \quad (F \in \mathcal{E}).$$

Proposition 16 (The angle detects modularity)

Let $F \in \mathcal{E}_k$ *and write* $\mathcal{G}_F$ *for* $\mathcal{G}_{\mathrm{Lat}(F)}$. *Then*

- F *is modular* $\iff$ $\mathcal{G}_F(\tau, \phi) = e^{-ik\phi} F(\tau)$,
- F *is quasimodular* $\iff$ $e^{ik\phi}\mathcal{G}_F(\tau, \phi)$ *is a polynomial in* $e^{2i\phi}$.

Backup: the angle at work

Example 17 (A cusp form made of double MES)

$F = 28\ \mathrm{G}_{9,3} + 150\ \mathrm{G}_{7,5} + 168\ \mathrm{G}_{5,7} - \frac{5197}{691}\ \mathrm{G}_{12}$ has a flat angle profile, so F is modular (Proposition 16). Its constant term vanishes, so $F = -\Delta/221120$ (Gangl-Kaneko-Zagier (2006)).